# MKQA: A Linguistically Diverse Benchmark for Multilingual Open Domain Question Answering

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## Abstract

Progress in cross-lingual modeling depends on challenging, realistic, and diverse evaluation sets. We introduce Multilingual Knowledge Questions and Answers (MKQA), an open-domain question answering evaluation set comprising 10k question-answer pairs aligned across 26 typologically diverse languages (260k question-answer pairs in total). Answers are based on heavily curated, language-independent data representation, making results comparable across languages and independent of language-specific passages. With 26 languages, this dataset supplies the widest range of languages to-date for evaluating question answering. We benchmark a variety of state-of-the-art methods and baselines for generative and extractive question answering, trained on Natural Questions, in zero shot and translation settings. Results indicate this dataset is challenging even in English, but especially in low-resource languages.[1]

## 1 Introduction

Training and evaluation data for question answering (QA) is severely lacking outside of high-resource languages like English. As unsupervised, transfer learning and zero/few-shot methods narrow the multilingual performance gap with English (Conneau et al., 2020; Lee and Lee, 2019; Cui et al., 2019a; Lewis et al., 2020), their real progress is hard to measure without challenging, realistic, and linguistically diverse evaluation sets. Existing multilingual QA datasets are realistic and challenging, but they lack linguistic diversity, comparable evaluation between languages, and are often limited to passages provided with the dataset (see Table 2).

We introduce Multilingual Knowledge Questions and Answers (MKQA) for evaluation of open-domain question answering. MKQA selects 10k realistic English queries from the Natural Questions dataset (NQ, Kwiatkowski et al., 2019) and human translates them into 25 additional languages and dialects. Accompanying these query translations we replace NQ's passage embedded answer spans with high-quality, language- and retrieval-independent answer annotations, linked directly against Wikidata entities and a limited set of well-defined value types (numbers, dates, strings, etc.).[2]

See one full example in Table 1. More flexible than existing multilingual datasets, MKQA's grading procedure ensures these labels are sufficient to evaluate any QA method, including knowledge graph and generative approaches. The objective of this evaluation set is to facilitate fair comparison between languages, without imposing assumptions on the underlying QA approach. We see MKQA as a useful tool enabling practitioners to benchmark a variety of multilingual open domain question answering methods against the widest range of available languages yet. Below, we discuss its central properties as an evaluation benchmark.

**Realistic & Reliable Annotations** Of crucial importance to any evaluation set is (a) how well it reflects realistic, real-world settings, and (b) the reliability of its annotations. To ensure the English queries, which form the basis of our dataset, are realistic, we use Natural Questions, formulated by real users, independent of passages or answers. To ensure these queries are realistic in other languages we employ expert bilingual translators, guided by strict localization criteria. We confirm that a large majority of these queries are **geographically invariant**, meaning that their answer is not culturally or geographically dependent (we found that less than 4% of answers are rendered incorrect by geographical and cultural context, for more details

[1] MKQA data and evaluation scripts are available at https://github.com/apple/ml-mkqa.

[2] Wikidata is a collaboratively edited open knowledge graph: https://www.wikidata.org/.

| Language | Language Code | Question | Answer and Aliases | Answer Type | Entity ID |
|---|---|---|---|---|---|
| English | | who produces the most pistachios in the world | Iran, Republic of Iran, ir, Islamic Republic of Iran, Persia, Islamic Rep. Iran | Entity | Q794 |
| Arabic | ar-SA | من أكبر منتج للفستق في العالم | إيران, ايران, الجمهورية الإسلامية الإيرانية, ايران, الجمهورية الإسلامية في إيران, الجمهور الإسلامية الإيرانية, جمهورية إيران الإسلامية, جمهورية إيران الإسلامية, جمهورية إيران, الجمهورية الاسلامية الايرانية, جمهوريّة إيران الإسلامية, الجمهورية الإسلامية الايرانية, ايران الأكبر | | |
| Danish | da-DK | hvem producere flest pistacienødder i verden | Iran | | |
| German | de-{DE,AT,CH} | Wer produziert weltweit die meisten Pistazien | Iran, Islamische Republik Iran | | |
| Spanish | es-ES | ¿Quién produce la mayor cantidad de pistachos del mundo? | Irán, República Islámica de Irán, Iran, Republica Islamica de Iran | | |
| Finnish | fi-FI | Kuka tuottaa eniten pistaasipähkinöitä maailmassa | Iran | | |
| French | fr-FR | Qui produit le plus de pistaches dans le monde | Iran, Perse, République islamique d'Iran | | |
| Hebrew | he-IL | מי מייצר הכי הרבה פיסטוקים בעולם | איראן | | |
| Hungarian | hu-HU | Ki termeli a legtöbb pisztáciát a világon? | Irán, Iráni Iszlám Köztársaság, Perzsia | | |
| Italian | it-IT | Chi produce più pistacchi al mondo | Iran, Repubblica Islamica dell'Iran | | |
| Japanese | ja-JP | 世界で一番ピスタチオを生産しているのは誰ですか | イラン, イラン　イスラム共和国 | | |
| Khmer | km-KH | អ្នកណាផលិត pistachios ច្រើនជាងគេបំផុតនៅលើពិភពលោក? | អ៊ីរ៉ង់ | | |
| Korean | ko-KR | 전 세계에서 누가 가장 많은 피스타치오를 생산하나요 | 이란 | | |
| Malay | ms-MY | siapa menghasilkan pistachios paling banyak di dunia | Iran | | |
| Dutch | nl-NL | wie produceert de meeste pistachio nootjes ter wereld | Iran, Islamitische Republiek Iran, IR, IRN | | |
| Norwegian | nb-NO | hvem lager mest pistasjnøtter i verden | Iran | | |
| Polish | pl-PL | kto produkuje najwięcej pistacji na świecie | Iran | | |
| Portuguese | pt-BR | quem produz mais pistaches no mundo | Irão, ايران, República Islâmica do Irão, Republica Islamica do Irao | | |
| Russian | ru-RU | кто в мире производит больше всего фисташек | Иран, Исламская Республика Иран, Персия, ИРИ, Iran, Иран (Исламская Республика), Название Ирана | | |
| Swedish | sv-SE | Vem producerar mest pistagenötter i världen? | Iran | | |
| Thai | th-TH | ใครผลิตถั่วพิตาชิโอมากที่สุดในโลก | ประเทศอิหร่าน | | |
| Turkish | tr-TR | Dünyanın en çok antep fıstığını kim üretiyor? | İran, İran İslam Cumhuriyeti | | |
| Vietnamese | vi-VN | nước nào sản xuất nhiều quả hồ trăn nhất thế giới | Iran | | |
| Chinese (Simplified) | zh-CN | 谁生产了世界上最多的开心果 | 伊朗 | | |
| Chinese (Hong Kong) | zh-HK | 誰是世界上開心果產量最高的人 | 伊朗, 伊朗伊斯蘭共和國 | | |
| Chinese (Traditional) | zh-TW | 全世界開心果的主要生產者是誰 | 伊朗 | | |

Table 1: **Questions and answers in all supported languages for one instance in MKQA.** The IETF BCP-47 language codes specify the language and locale. The Entity ID corresponds to Wikidata (see for instance `https://www.wikidata.org/wiki/Q794`).

see Section 4.2). To ensure annotation reliability, we enforce minimum inter-grader agreement, conduct quality checks, and re-annotation from expert graders where necessary. Further, the Wikidata entity identifiers (QIDs) ground the answer annotations in structured data. This can be used for other knowledge graph-specific metrics, to retrieve other valid answer strings, and trivial entity translation into hundreds of languages beyond the scope of MKQA.

**Parallel Questions** Our evaluation set is fully aligned, or "parallel", across all available languages, meaning the same examples exist in all languages. This is accomplished by a mixture of expert human translation and using multilingual data from Wikidata. This property enables direct comparison between all 26 languages for fully cross-lingual or zero-shot systems. While Clark et al. (2020) point out the natural query distribution varies by language and geography, we reserve our assessment to geographically invariant queries for the purpose of more fair comparison between methods.

**Retrieval-Independent Annotations** Existing training and evaluation sets are oriented to "extractive" QA, providing specific passages and passage-dependent answer annotations (Clark et al., 2020; Lewis et al., 2020; Artetxe et al., 2020b; Liu et al., 2019a). These types of annotations are of limited use with varying retrieval systems, knowledge graph approaches, and even generative approaches because the answers are tied to the particular phrasing of their passage. Translating annotations from English passages may also introduce "translationese artifacts" as the translation is implicitly influenced by the original English structure (Artetxe et al., 2020a). These artifacts render the task easier for methods relying on English supervision or machine translation techniques. As we shall discuss in Section 3, the MKQA collection procedure yields primarily entity and structured "atomic" answer types. We contend retrieval-independent (and particularly entity-oriented) annotations minimize the risk of translation artifacts, and remove limitations on the underlying QA approach.

**Linguistic Diversity** Lastly, MKQA has broad linguistic diversity, covering 26 languages and dialects from 14 language family branches. Languages from MKQA cover half of the world populations' native language, and more than 90% of the world population lives in a country where one of these languages is an official language (see Section 4.1 for more details). It is to our knowledge both the largest and most linguistically diverse open-domain QA evaluation set currently available (see Tables 2 and 3).

MKQA makes two important contributions to the field of multilingual question answering:

- Our *answer collection procedure* renders the evaluation set highly reliable, independent, and unbiased towards the QA technique used.

| Multilingual QA Evaluation Set | Answer Independence | Parallel Questions | Language Fam. Branches | Languages | Total Examples |
|---|---|---|---|---|---|
| XQA (Liu et al., 2019a) | ✓ | × | 5 | 9 | $28k$ |
| MLQA (Lewis et al., 2020) | × | ✓ | 6 | 7 | $46k$ |
| XQuAD (Artetxe et al., 2020b) | × | ✓ | 11 | 11 | $13k$ |
| TyDi (Clark et al., 2020) | × | × | 11 | 11 | $204k$ |
| Xor-QA (Asai et al., 2021) | × | × | 7 | 7 | $40k$ |
| MKQA (This work) | ✓ | ✓ | 14 | 26 | $260k$ |

Table 2: **Comparison of multilingual QA evaluation sets.** Answer independence indicates whether the gold answer is independent of a retrieved document, and parallel questions indicates whether examples are the same across languages.

This unique setup allows us to fairly compare the performance of techniques as distinct as knowledge graph-based, dense and sparse retrieval and generative QA techniques on a large number of languages (see Section 5).

- Our dataset provides *fully aligned* examples in the largest yet number of typologically diverse languages, enabling comparable evaluation across many languages.

We find MKQA is innately more challenging than Natural Questions from which it was derived, due to the multi-stage re-annotation process. The best model obtains only 52.3% F1 in English, and only 5.7% above a naive baseline on the lowest resource language. Given these qualities, our dataset facilitates broad and reliable evaluation of multilingual, open-domain question answering.

## 2 Related Work

**Cross-Lingual Modeling** Recent work trains cross-lingual representations with unsupervised language modeling over many languages, including Multilingual BERT (Devlin et al., 2019), XLM-R (Conneau et al., 2020), and Multilingual T5 (Xue et al., 2021). Transfer learning techniques are often applied to these cross-lingual representations to overcome the dearth of non-English data (Cui et al., 2019a; Hsu et al., 2019; Lee and Lee, 2019; Kumar et al., 2019). Recent investigations into cross-lingual modeling have revealed "translation artifacts" in datasets where machine translation systems are used, or human translation tasks are not carefully curated (Artetxe et al., 2020a; Wintner, 2016; Rabinovich and Wintner, 2015). "Translationese" results in hidden linguistic cues in translated text that render the task easier than a natural translation.

**English QA Resources** A majority of question answering research focuses on English, which offers ample selection of evaluation datasets, including SQuAD (Rajpurkar et al., 2016), TriviaQA (Joshi et al., 2017), and Natural Questions (Kwiatkowski et al., 2019). Open Domain QA, pioneered by Green et al. (1986), is the task of answering open questions using external knowledge sources. A common approach is to combine retrieval and extractive techniques (Chen et al., 2016, 2017; Dhingra et al., 2017; Cui et al., 2017).

**Monolingual QA Resources** Non-English question answering resource options remain comparatively rare, with most options spanning only one other language, and rarely low-resource languages. DuReader (He et al., 2018), CMRC (Cui et al., 2019b), and DRCD (Shao et al., 2018) all offer high-quality Chinese QA datsets. Similarly, XCMRC (Liu et al., 2019b) and BiPar (Jing et al., 2019) present parallel, cross-lingual QA dataset between English and Chinese. Exploring slightly less resource-rich languages, numerous works have derived new datasets from SQuAD, employing varying degrees of human or semi-automatic translation techniques to non-English target languages: ARCD for Arabic (Mozannar et al., 2019), KorQuAD-1.0 for Korean (Lim et al., 2019), and MMQA for Hindi (Gupta et al., 2018).

**Multilingual QA Resources** Table 2 compares the largest publicly available multilingual question answering evaluation sets. The table highlights the following properties of each dataset: whether the available gold answers are independent of retrieved documents, whether examples are aligned across languages, and the number of languages and examples provided. MLQA (Lewis et al., 2020) and XQuAD (Artetxe et al., 2020b) are examples of SQuAD-style extractive datasets, employing human translators to create parallel examples. Both MLQA and XQuAD ensure that all answers are answerable (discarding "No Answer" exam-